STAFF REVIEW · H446 268 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Eligible for CPU time.

2. Maximum CPU time per RR turn.

3. Saving and restoring execution context when changing process.

4. mid4→9, mid6→13, mid7→15 found; indexes 4,6,7.

I Do — source worked steps

1. Predict first — Shorter job B runs first

2. Model and check — A waits two time units.

We Do — guided CHECK

1. Predict first
Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.
Model and check
Explain why each value follows from the earlier step.

You Do — source mark schemes

1. Arrival order [1]; shortest available non-preemptive job [1]; preemptive shortest remaining [1]; fixed time quanta [1]; adaptive multiple queues [1].

2. FCFS A,B,C [1], waits 0,5,8 [1], mean13/3 [1]; SJF C,B,A [1], waits 0,1,4 [1], mean5/3 [1].

3. A0–2 [1], B2–4 [1], C4–5 [1], A5–7 [1], B7–8 [1]; small quantum adds context-switch overhead [1].

Exit Ticket — source answer

1. False. It normally preempts each process after a time quantum.